A Universal Beta Formula for Gamma Normal Forms

All coefficients, exact support and signs,
endpoint primitives, and polylogarithmic–Stieltjes transforms

Research continuation prepared for Vladimir Reshetnikov

Mathematical derivations and computational checks with OpenAI assistance

11 October 2026

Abstract. We give a finite rational formula for every coefficient in the canonical integration-by-parts normal form of $(-1)^m\Gamma(x)(\Gamma(x)^{-1})^{(m)}$. Each coefficient is one polynomial beta integral. The formula determines its exact support: the previously known factor-parity condition is also sufficient, and every nonzero coefficient has the sign of the parity of its highest derivative index. Thus the mixed-square question in the preceding affine-polarization report is solved for all orders, as are the higher-multiplicity and coupled sectors of this particular Gamma hierarchy.

The proof transports a constant-coefficient quadratic identity to every polynomial background by an auxiliary differential operator. The highest quadratic variation then determines the full normal form, not only its quadratic sector. A finite entire generating function and an explicit Euler-homotopy primitive retain all integration endpoints. The resulting polynomial weights give finite Hurwitz and Lerch endpoint formulas, generalized-harmonic logarithmic moments, all-order Stieltjes averages and anchored primitives, and finite polylogarithmic evaluations. A degree-eight weight yields explicit trigamma, γ_0, γ_1, and log-sine identities.

The classical normal-form complement, the affine square kernel, and the standard special-function calculus are credited to their sources. The all-order proofs are separate from 8,350 independent rational coefficient comparisons through weight 32 and 20 floating-point analytic diagnostics. Historical priority beyond the inspected sources is not asserted. No proof-assistant formalization or interval certification is claimed.

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1 The continuation problem and the universal formula

The immediate predecessor, *Affine Polarization of Gamma Normal Forms* [3], determines every coefficient of $u_0^a u_1^k u_r^2$. Its first further-research question asks for the coupled family $u_0^a u_1^k u_2^b u_3^2$, with $b > 0$; its second asks for nonlinear polarization and higher multiplicities. The earlier *Exact Harmonic-Tail Reconstruction and Variational Gamma Identities* [2] gives the complete u_0, u_1, u_2 projection through a variational recurrence. We solve these coefficient questions in a stronger form: every functional monomial of the ordinary inverse-Gamma hierarchy has an explicit finite formula.

This is a theorem about a specified rational differential-polynomial normal form. It does not assert that arbitrary individual polygamma integrals have elementary antiderivatives, or that a set of zeta values is arithmetically independent. The original Gaussian S_6, revised S_8, and general unequal harmonic-tail moment-prefix questions are not settled here.

Let

$$\mathscr{R} = \mathbb{Q}[u_0, u_1, u_2, \ldots], \qquad \mathcal{D}u_j = u_{j+1}, \qquad \mathrm{wt}(u_j) = j + 1. \tag{1.1}$$

A monomial is *functional* if it is a power of u_0, including 1, or if its largest positive index occurs at least twice. Write $\mathcal{N}$ for the unique projection onto the rational span of these monomials, along total derivatives. Its construction and uniqueness are recalled in Section 2. Define

$$A_0 = 1, \qquad A_{m+1} = u_0 A_m - \mathcal{D}A_m, \qquad N_m = \mathcal{N}(A_m). \tag{1.2}$$

Under $u_j = \psi^{(j)}(x)$, with $\psi^{(0)} = \psi$, one has

$$A_m(\psi(x), \psi'(x), \ldots) = (-1)^m \Gamma(x) \frac{\mathrm{d}^m}{\mathrm{d}x^m} \frac{1}{\Gamma(x)}. \tag{1.3}$$

For a polynomial $Q(v)$ put

$$\langle Q \rangle = \frac{1}{2} \int_{-1}^{1} Q(v)\, \mathrm{d}v, \qquad \Theta_j(v) = \frac{(v+1)^{j+1} - (v-1)^{j+1}}{2^{j+1}(j+1)!} \quad (j \geq 1). \tag{1.4}$$

The first three polynomials are

$$\Theta_1 = \frac{v}{2}, \qquad \Theta_2 = \frac{1 + 3v^2}{24}, \qquad \Theta_3 = \frac{v + v^3}{48}. \tag{1.5}$$

All nonzero coefficients of Θ_j are positive, and $\Theta_j(-v) = (-1)^j \Theta_j(v)$.

Theorem 1.1 (Every functional coefficient)**.** *Let $d \geq 1$, $a, e_1, \ldots, e_d \geq 0$, and $e_d \geq 2$. Put*

$$M = u_0^a \prod_{j=1}^{d} u_j^{e_j}, \qquad m = a + \sum_{j=1}^{d} (j+1) e_j,$$

and define the rational polynomial

$$P_M(v) = (1 - v^2)^d \Theta_d(v)^{e_d - 2} \prod_{j=1}^{d-1} \Theta_j(v)^{e_j}. \tag{1.6}$$

Then

$$\boxed{[M] N_m = \frac{(-1)^d m!}{2^{2d} d!(d+1)!\, a! \prod_{j=1}^{d} e_j!} \, \langle P_M \rangle \,.} \tag{1.7}$$

The coefficient of u_0^m is 1. A monomial with a unique largest positive index has coefficient zero. Coefficients at a different total weight are zero.

The average in (1.7) is a finite rational operation: if $P_M(v) = \sum q_k v^k$, then $\langle P_M \rangle = \sum_{k \text{ even}} q_k/(k+1)$. Thus the theorem is not an implicit recurrence or a numerical integral prescription. It avoids constructing the full Bell polynomial or reducing all of its partitions.

Corollary 1.2 (Exact support and sign)**.** *For the functional monomial in Theorem 1.1,*

$$[M]N_m = 0 \quad \Longleftrightarrow \quad \sum_{j=1}^{d} j e_j \text{ is odd.} \tag{1.8}$$

When the coefficient is nonzero, its sign is exactly $(-1)^d$*.*

Proof. The integrand has parity $\sum j e_j - 2d$, which is the parity in (1.8). In the even case it is strictly positive for $0 < v < 1$, since every $\Theta_j(v)$ is positive there. The prefactor in (1.7) has sign $(-1)^d$. □

The parity condition alone was already available from centering [1, 2, 3]. Its sufficiency and the uniform sign law follow from the new coefficient formula. In particular, there are no additional accidental zero coefficients within this functional basis.

1.1 A positive finite-sum version

Suppose the parity is even and write

$$R_M(v) = \Theta_d(v)^{e_d - 2} \prod_{j<d} \Theta_j(v)^{e_j} = \sum_{\ell \geq 0} r_\ell v^{2\ell}.$$

Every $r_\ell \geq 0$, and only finitely many are nonzero. The beta integral

$$\int_0^1 v^k (1-v^2)^d \, \mathrm{d}v = \frac{2^d d!}{\prod_{h=0}^{d} (k+2h+1)}, \qquad \Re k > -1, \tag{1.9}$$

gives the equivalent formula

$$[M]N_m = \frac{(-1)^d m!}{2^d (d+1)!\, a! \prod_j e_j!} \sum_{\ell \geq 0} \frac{r_\ell}{\prod_{h=0}^{d} (2\ell + 2h + 1)}. \tag{1.10}$$

This representation does not subtract alternating coefficients of $(1-v^2)^d$. It also makes the nonvanishing transparent.

2 Canonical reduction and the centered Gamma series

The complement used here is classical in differential algebra [4, Section 4.4.1] and is already established in the ProveIt manuscript [1]. We include the elementary proof because an exact coefficient theorem depends on the precise choice of complement.

Lemma 2.1 (Functional decomposition)**.** *Every $P \in \mathscr{R}$ has a unique decomposition*

$$P = \mathcal{N}(P) + \mathcal{D}\mathcal{H}(P), \qquad \mathcal{N}(P) \text{ functional}, \qquad [1]\mathcal{H}(P) = 0. \tag{2.1}$$

Both maps preserve factor degree. On a homogeneous polynomial of positive weight m, the remainder has weight m and the primitive has weight $m-1$.

Proof. If a monomial has unique largest index $r \geq 1$, write it as $u_r u_{r-1}^a V$, where V uses indices below $r-1$. Then

$$u_r u_{r-1}^a V = \mathcal{D}\left(\frac{u_{r-1}^{a+1} V}{a+1}\right) - \frac{u_{r-1}^{a+1} \mathcal{D} V}{a+1}. \tag{2.2}$$

Every monomial of the remainder has largest index below r. Repeated replacement terminates. The stated gradings are preserved. For uniqueness, let a nonconstant H have largest variable index j. The terms $u_{j+1}\partial H/\partial u_j$ in $\mathcal{D}H$ are nonzero, have largest index $j+1$ exactly once, and cannot cancel terms of smaller largest index. Hence $\mathcal{D}H$ cannot be functional. The only remaining freedom in H is a scalar, removed by the normalization. □

The quotient is a vector-space quotient, not an algebra quotient. For example, $\mathcal{N}(u_1) = 0$ but $\mathcal{N}(u_1^2) = u_1^2$. Nor may one truncate high derivatives before reduction: $\mathcal{N}(u_0 u_4) = u_2^2$. All restrictions to finitely many jet variables below are performed *after* applying $\mathcal{N}$.

The exponential generating function of (1.2) is

$$\mathcal{A}(t) = \sum_{m \geq 0} A_m \frac{t^m}{m!} = \exp\left(\sum_{j \geq 0} \frac{(-1)^j u_j t^{j+1}}{(j+1)!}\right). \tag{2.3}$$

Its centered translate is

$$\mathcal{C}(t) = e^{t\mathcal{D}/2}\mathcal{A}(t) = \exp\left(\sum_{j \geq 0} \frac{u_{2j} t^{2j+1}}{2^{2j}(2j+1)!}\right), \tag{2.4}$$

$$\mathcal{N}\mathcal{C}(t) = \mathcal{N}\mathcal{A}(t) = \sum_{m \geq 0} N_m \frac{t^m}{m!}. \tag{2.5}$$

Indeed, $(e^{t\mathcal{D}/2} - 1)P(t)$ is a total derivative coefficientwise. Under the digamma substitution, (2.3) is $\Gamma(x)/\Gamma(x-t)$, while (2.4) is $\Gamma(x + t/2)/\Gamma(x - t/2)$ near $t = 0$ away from poles.

Translation $u_0 \mapsto u_0 + c$ commutes with $\mathcal{D}$ and preserves the functional span, so it commutes with $\mathcal{N}$. Consequently

$$\partial_{u_0}\mathcal{N}\mathcal{C}(t) = t\mathcal{N}\mathcal{C}(t), \qquad \partial_{u_0} N_m = m N_{m-1}. \tag{2.6}$$

Let σ_d set $u_j = 0$ for $j > d$, and put $z_j = u_j t^{j+1}$. The weight grading and (2.6) give unique series F_d with

$$\sigma_d \mathcal{N}\mathcal{C}(t) = e^{tu_0} F_d(z_1, \ldots, z_d), \qquad F_0 = 1. \tag{2.7}$$

Normal support implies

$$F_d|_{z_d=0} = F_{d-1}, \qquad \partial_{z_d} F_d|_{z_d=0} = 0. \tag{2.8}$$

The problem is therefore to determine the second derivative of F_d with respect to its highest jet coordinate. This is precisely what quadratic variation on a polynomial background accomplishes.

3 Weighted quadratic forms and the affine kernel

We first recover the affine square kernel of [3]. This part is an inherited input, reproduced to fix its normalization and to expose the transport operation used in the next section. Let

$$K_t(\xi) = \frac{2\sinh(t\xi/2)}{\xi}, \qquad K_t(0) = t. \tag{3.1}$$

The centered series is formally $\mathcal{C}(t) = \exp(K_t(\mathcal{D})u_0)$. For a perturbation $u_0 = q + px + \varepsilon h(x)$, the quadratic coefficient is

$$[\varepsilon^2]\mathcal{C}(t) = \frac{1}{2}e^{tq+tpx}(K_t(\partial_x)h)^2. \tag{3.2}$$

Lemma 3.1 (Diagonal uniqueness with variable coefficients)**.** *A finite local quadratic differential expression in h can be reduced, modulo an ordinary total x-derivative, to a sum $\sum_r a_r(x)(h^{(r)})^2$. The diagonal coefficients are unique. If a term before reduction is $a(x)h^{(i)}h^{(j)}$, no square of order greater than $\lfloor (i+j)/2 \rfloor$ is needed. These statements also hold coefficientwise for formal t-series with finite differential order at every coefficient.*

Proof. Integration by parts moves a derivative from the larger index to the smaller one or to the coefficient. The former reduces the difference of indices, and the latter reduces their sum. Terms of neighboring orders are reduced by $2h^{(r)}h^{(r+1)} = ((h^{(r)})^2)'$. This proves existence and the order assertion.

For uniqueness, suppose $\sum_{r=0}^{R} a_r(x)(h^{(r)})^2$ is a local total derivative. Its Euler derivative with respect to h is zero. The coefficient of $h^{(2R)}$ in that Euler derivative is $2(-1)^R a_R(x)$. Thus $a_R = 0$, and descending induction gives all $a_r = 0$. Explicit x-dependence of the coefficients causes no problem. The primitive must be a finite-order local differential expression; an unrestricted nonlocal antiderivative is not part of this quotient. □

Lemma 3.2 (Affine diagonal kernel)**.** *Let κ be an auxiliary constant, independent of t. Modulo local total derivatives,*

$$\frac{1}{2}e^{\kappa x}(K_t(\partial_x)h)^2 \equiv e^{\kappa x}\sum_{r\geq 0} t^{2r+2} g_r(t\kappa/2)(h^{(r)})^2, \tag{3.3}$$

where

$$c_r = \frac{(-1)^r}{2^{2r+1}r!(r+1)!}, \tag{3.4}$$

$$g_r(y) = c_r\left\langle (1-v^2)^r e^{yv}\right\rangle = c_r\int_0^1 (1-v^2)^r\cosh(yv)\,\mathrm{d}v. \tag{3.5}$$

Every coefficient of the primitive in (3.3) can be chosen as $e^{\kappa x}$ times a quadratic differential polynomial with coefficients in $\mathbb{Q}[\kappa]$.

Proof. Polarize a quadratic expression using derivative symbols ξ_1, ξ_2. A weighted derivative has symbol divisible by $\xi_1 + \xi_2 + \kappa$. In the quotient put $\xi_1 = \xi$, $\xi_2 = -\xi - \kappa$, and $\eta = -\xi(\xi + \kappa)$. The symbol of $(h^{(r)})^2$ is η^r. Symmetric polynomials subject to this linear relation form the polynomial ring in η over $\mathbb{Q}[\kappa]$.

With $y = t\kappa/2$, the product-to-sum identity gives

$$\begin{aligned}\frac{1}{2}K_t(\xi)K_t(-\xi-\kappa) &= \frac{\cosh y - \cosh\sqrt{y^2 - t^2\eta}}{\eta}\\ &= \sum_{r\geq 0}\frac{(-1)^r t^{2r+2}}{(r+1)!}\left(\partial_{y^2}^{r+1}\cosh y\right)\eta^r.\end{aligned} \tag{3.6}$$

The entire series $\cosh\sqrt{z} = \sum_{n\geq 0} z^n/(2n)!$ introduces no branch, and the numerator is divisible by η. The beta integral (1.9), compared coefficientwise with the last line, gives exactly (3.5). At every power of t, the difference of the two quadratic symbols is divisible by $\xi_1 + \xi_2 + \kappa$. Polynomial division gives the asserted local primitive without introducing denominators in κ. □

The indices $r = 0, 1$ in (3.3) are diagonal *quadratic-variation* coefficients. They are not the constant and linear coefficients of a normal-form generating function. Confusing these meanings would destroy the highest-variation argument below.

4 Polynomial-background transport and the all-order proof

The essential new operation is simple: differential superposition in the auxiliary affine slope transports the *entire local identity*, including its primitive, to an arbitrary polynomial weight. It is not a substitution of a variable slope into a constant-coefficient quotient.

Lemma 4.1 (Differential superposition)**.** *Let $P_t(X)$ be a polynomial in X with formal t-series coefficients, with $P_t(X) = O(t)$. For any constant a, coefficientwise in t,*

$$e^{P_t(\partial_\kappa)} e^{\kappa(x+a)}\Big|_{\kappa=0} = e^{P_t(x+a)}. \tag{4.1}$$

The operation commutes with ∂_x and hence transports local integration-by-parts identities to local integration-by-parts identities.

Proof. The exponential $e^{\kappa(x+a)}$ is an eigenfunction of every polynomial in ∂_κ. Expanding the outer exponential proves (4.1). Because $P_t = O(t)$, only finitely many powers of that operator are relevant at a specified coefficient of t. The operators ∂_x and ∂_κ commute. Locality in the perturbation and finite differential order at each coefficient are preserved. The assertion remains valid with $a = tv/2$ and subsequent integration over the compact parameter interval $-1 \le v \le 1$. □

Let $b(x)$ be an arbitrary polynomial background, and define its centered integral

$$P_t(x) = \int_{-t/2}^{t/2} b(x+s)\,\mathrm{d}s. \tag{4.2}$$

The integral here means its finite polynomial expansion, so no analytic assumption on b is necessary. For $u_0 = b(x) + \varepsilon h(x)$,

$$[\varepsilon^2]\mathcal{C}(t) = \frac{1}{2} e^{P_t(x)} (K_t(\partial_x)h)^2. \tag{4.3}$$

Taylor expansion at x gives the key identity

$$P_t(x + tv/2) = tb(x) + \sum_{j\ge 1} b^{(j)}(x) t^{j+1} \Theta_j(v). \tag{4.4}$$

Indeed, the coefficient of $b^{(j)}(x)$ is $j!^{-1} \int_{-t/2}^{t/2} (tv/2 + s)^j \,\mathrm{d}s$, which is $t^{j+1}\Theta_j(v)$.

Theorem 4.2 (Polynomial-background quadratic kernel)**.** *If b has degree at most d and $z_j = b^{(j)}(x)t^{j+1}$, the unique diagonal form of (4.3) is*

$$e^{tb(x)} \sum_{r\ge 0} t^{2r+2} G_r(z_1, \dots, z_d)(h^{(r)})^2, \tag{4.5}$$

where

$$\boxed{G_r(\boldsymbol{z}) = c_r \left\langle (1-v^2)^r \exp\left(\sum_{j=1}^{d} z_j \Theta_j(v) \right) \right\rangle}. \tag{4.6}$$

Each G_r is an entire function of the finitely many z_j.

Proof. Apply $e^{P_t(\partial_\kappa)}$ to the affine local identity (3.3), and evaluate at $\kappa = 0$. Its left side becomes (4.3). On the right, write the factor $e^{\kappa x} g_r(t\kappa/2)$ as

$$c_r \left\langle (1-v^2)^r e^{\kappa(x+tv/2)} \right\rangle .$$

Lemma 4.1 sends this to $c_r \left\langle (1-v^2)^r e^{P_t(x+tv/2)} \right\rangle$. Now use (4.4). The transported primitive is local, and diagonal uniqueness is supplied by Lemma 3.1. Compact integration of an exponential of a polynomial proves entire dependence on the finite parameter list. □

Proposition 4.3 (The highest-variation equation)**.** *For every $d \geq 1$, the canonical generating function* (2.7) *satisfies*

$$\boxed{\partial_{z_d}^2 F_d(z_1, \ldots, z_d) = 2G_d(z_1, \ldots, z_d).} \tag{4.7}$$

Proof. Take a polynomial b of degree at most d, but perturb it by an arbitrary h; do not truncate $h^{(j)}$ for $j > d$. The identity $\mathcal{C} - \mathcal{N}\mathcal{C} = \mathcal{D}H$ implies equality of the diagonal quadratic variations of the two sides after substitution.

First consider a normal monomial with largest index $r > d$. Its highest factor must occur at least twice. At order ε^2 it contributes only if it contains exactly two factors above d, both of them u_r. Its contribution is already a diagonal square $(h^{(r)})^2$ of order $r > d$. Monomials with three or more factors above d do not contribute to quadratic variation at all.

For a monomial involving only $u_0, \ldots, u_d$, the second variation contains terms $h^{(i)}h^{(j)}$ with $i, j \leq d$. Unless $i = j = d$, Lemma 3.1 shows that its diagonal reduction has order at most $d-1$. Thus the coefficient of the square $(h^{(d)})^2$ in the diagonal variation of the normal form is precisely

$$\frac{1}{2}\partial_{u_d}^2 (e^{tu_0} F_d) = \frac{1}{2} e^{tu_0} t^{2d+2} \partial_{z_d}^2 F_d.$$

Theorem 4.2 gives the coefficient on the original centered density as $e^{tu_0} t^{2d+2} G_d$. Diagonal uniqueness proves the equality after every polynomial-background substitution. At a fixed x, arbitrary values of $b(x), \ldots, b^{(d)}(x)$ are realized by a polynomial of degree d. Hence this is an identity in the independent jet variables, proving (4.7). □

Proof of Theorem 1.1. By (4.7), the coefficient of $z_1^{e_1} \cdots z_d^{e_d}$ in F_d, for $e_d \geq 2$, is

$$\frac{2c_d}{\prod_{j=1}^d e_j!} \left\langle (1-v^2)^d \Theta_d(v)^{e_d - 2} \prod_{j<d} \Theta_j(v)^{e_j} \right\rangle .$$

Multiply by the coefficient $1/a!$ from e^{tu_0} and by $m!$ from the exponential generating convention. Since $2c_d = (-1)^d/(2^{2d} d!(d+1)!)$, this is (1.7). The initial conditions (2.8) supply the remaining highest-index cases, and $F_0 = 1$ supplies the pure powers of u_0. □

This proof is all-order. The finite rational comparisons reported later are independent checks of normalizations and implementation, not the reason the coefficient theorem holds for untested weights.

5 Entire finite-jet generators and solved coefficient families

5.1 A recursion-free generating function

Define the entire divided difference

$$E_2(z,w)=\frac{e^{zw}-1-zw}{w^2}=\sum_{n\ge 2}\frac{z^n w^{n-2}}{n!},\qquad E_2(z,0)=\frac{z^2}{2}. \tag{5.1}$$

Two integrations of the highest-variation equation, using (2.8), give the following complete generator.

Corollary 5.1 (Finite-jet generating function)**.** *For every $d\ge 0$,*

$$\boxed{F_d=1+\sum_{r=1}^{d}2c_r\left\langle (1-v^2)^r\exp\left(\sum_{j<r}z_j\Theta_j(v)\right)E_2(z_r,\Theta_r(v))\right\rangle.} \tag{5.2}$$

It is entire in $(z_1,\dots,z_d)$. All apparent divisions by zeros of Θ_r are removable. The infinite sum over r is also meaningful as a formal t-series after $z_j=u_jt^{j+1}$, because the rth summand starts at weight $2r+2$.

This formula closes the full finite-jet coefficient problem. In particular, the higher powers of the highest derivative are determined by integrating its quadratic variation twice in that coordinate. There is no need to compute a cubic or quartic symbol in several independent derivative variables. Such symbols remain useful for other hierarchies, but they are not an unresolved step for the coefficients in (1.2).

For $d=1$, write $y=z_1/2$. Equation (5.2) recovers the predecessor's low sector,

$$F_1(2y)=\frac{1}{2}\left(\cosh y+\frac{\sinh y}{y}-y\int_0^y\frac{\sinh w}{w}\,\mathrm{d}w\right), \tag{5.3}$$

with its removable value at the origin. For a quadratic background, (4.6) becomes the particularly simple heat identity

$$\begin{aligned}G_r(2y,z)&=e^{z/24}e^{(z/8)\partial_y^2}g_r(y)\\&=c_r\int_0^1(1-v^2)^re^{z(1+3v^2)/24}\cosh(yv)\,\mathrm{d}v.\end{aligned} \tag{5.4}$$

The scalar factor $e^{z/24}$ is essential: it is the constant term in the centered integral of the quadratic background. Dropping it would produce an incorrect mixed coefficient even at weight eleven.

5.2 The previously open mixed-square family

Corollary 5.2 (All lower quadratic powers with every higher square)**.** *Let $r\ge 3$ and $a,k,b\ge 0$, and put $m=a+2k+3b+2r+2$. For even k,*

$$\boxed{[u_0^a u_1^k u_2^b u_r^2]N_m=\frac{(-1)^r m!}{a!\,2^{k+r+1}k!(r+1)!\,b!24^b}\sum_{j=0}^{b}\frac{\binom{b}{j}3^j}{\prod_{\ell=0}^{r}(k+2j+2\ell+1)}.} \tag{5.5}$$

For odd k the coefficient is zero.

Proof. In (1.7), set $e_r = 2$, $e_1 = k$, $e_2 = b$, and all other exponents zero. Expand $(1+3v^2)^b$ and apply (1.9). The factor $1/2!$ for the highest square cancels the factor 2 in the highest-variation equation. □

At $b = 0$ this is the affine theorem of [3]. At $r = 3$ and every $b > 0$ it answers that report's first explicit coefficient question. The restriction $r \geq 3$ is important: when $r = 2$, the two displayed factors involving u_2 must be combined, changing their factorial normalization. The complete combined u_2 sector is instead the following formula.

Corollary 5.3 (A finite replacement for the three-variable recurrence)**.** *Let $b \geq 2$, $a, k \geq 0$, and $m = a + 2k + 3b$. For even k,*

$$[u_0^a u_1^k u_2^b] N_m = \frac{m!}{a!\, 2^k k!\, b! 24^{b-1}} \sum_{j=0}^{b-2} \frac{\binom{b-2}{j} 3^j}{(k+2j+1)(k+2j+3)(k+2j+5)}. \tag{5.6}$$

Odd k gives zero. This evaluates the all-u_2 family previously specified by the variational recurrence in [2, Section 4].

Proof. Use $d = 2$ in Theorem 1.1 and the same beta integral. □

5.3 Nonlinear sectors and explicit normal forms

The universal theorem is not confined to highest squares. For example,

$$\boxed{[u_1 u_3^3] N_{14} = -\frac{3185}{3}.} \tag{5.7}$$

Here $\Theta_1 \Theta_3 = (v^2 + v^4)/96$, so the calculation is a sum of two elementary beta integrals. In particular, the normal form is not in general integral over $\mathbb{Z}$, even though the original Bell polynomial A_m has integer coefficients. The first mixed-square values include

$$[u_2 u_3^2] N_{11} = -55, \qquad [u_2^2 u_3^2] N_{14} = -3640, \qquad [u_2 u_4^2] N_{13} = 91. \tag{5.8}$$

The entire low normal forms are useful normalization checks:

$$N_4 = u_0^4 - u_1^2, \tag{5.9}$$

$$N_8 = u_0^8 - 70 u_0^4 u_1^2 - 7 u_1^4 + 28 u_0^2 u_2^2 - u_3^2, \tag{5.10}$$

$$\begin{aligned} N_{11} = {} & u_0^{11} - 330 u_0^7 u_1^2 - 1155 u_0^3 u_1^4 + 462 u_0^5 u_2^2 + 990 u_0 u_1^2 u_2^2 \\ & + 550 u_0^2 u_2^3 - 165 u_0^3 u_3^2 - 55 u_2 u_3^2 + 11 u_0 u_4^2. \end{aligned} \tag{5.11}$$

These finite polynomials are not presented as historically new by themselves. The new assertion is the all-monomial formula which produces every such polynomial without an integration-by-parts search.

5.4 A support-counting identity

Let s_m be the number of nonzero monomials of N_m. The exact support theorem gives, as a formal power series,

$$\sum_{m \geq 0} s_m q^m = \frac{1}{1-q} \left\{ 1 + \frac{1}{2} \sum_{d \geq 1} q^{2d+2} \left[\prod_{j=1}^{d} \frac{1}{1-q^{j+1}} + \prod_{j=1}^{d} \frac{1}{1-(-1)^j q^{j+1}} \right] \right\}. \tag{5.12}$$

To prove it, reserve two copies of the highest variable u_d, allow arbitrary remaining nonnegative exponents below and at d, and apply the even-parity projector. The factor $(1-q)^{-1}$ accounts for u_0. Each coefficient involves only finitely many d. This counts support, not arithmetic relations among the values obtained after substituting polygamma functions.

6 Explicit primitives and Gamma–harmonic endpoint identities

The normal form is only half of an integration identity. We now give a finite formula for its retained endpoint primitive. Let

$$\mathscr{E}(R) = \sum_{r\geq 0} (-\mathcal{D})^r \frac{\partial R}{\partial u_r} \tag{6.1}$$

be the Euler derivative. All sums are finite for a fixed polynomial. The commutator $\partial_{u_r}\mathcal{D} = \mathcal{D}\partial_{u_r} + \partial_{u_{r-1}}$, with the last term omitted for $r=0$, gives $\mathscr{E}(\mathcal{D}H) = 0$ by cancellation.

Proposition 6.1 (An explicit normalized primitive)**.** *Let $R = \mathcal{D}H$ and decompose $R = \sum_{h\geq 1} R_h$ by factor degree. Then its unique constant-free primitive is*

$$\boxed{\mathcal{H}(R) = \sum_{h\geq 1} \frac{1}{h} \sum_{r\geq 1} \sum_{j=0}^{r-1} u_{r-1-j} (-\mathcal{D})^j \frac{\partial R_h}{\partial u_r}.} \tag{6.2}$$

In particular, with $R = A_m - N_m$, the same expression gives the primitive H_m in $A_m = N_m + \mathcal{D}H_m$.

Proof. Euler's identity for factor degree and repeated integration by parts give

$$\begin{aligned} hR_h &= \sum_{r\geq 0} u_r \frac{\partial R_h}{\partial u_r} \\ &= u_0 \mathscr{E}(R_h) + \mathcal{D}\left(\sum_{r\geq 1} \sum_{j=0}^{r-1} u_{r-1-j} (-\mathcal{D})^j \frac{\partial R_h}{\partial u_r} \right). \end{aligned}$$

Because $\mathcal{D}$ preserves factor degree, each R_h is itself a total derivative and its Euler derivative vanishes. Division by h proves the formula. Every displayed summand has positive factor degree, so there is no scalar term; uniqueness follows from Lemma 2.1. □

For $0 < \alpha < \beta$, substitution in the exact polynomial identity yields

$$\begin{aligned} &\int_\alpha^\beta (-1)^m \Gamma(x) \frac{\mathrm{d}^m}{\mathrm{d}x^m} \frac{1}{\Gamma(x)} \,\mathrm{d}x \\ &\qquad = \int_\alpha^\beta N_m(\psi(x), \psi'(x), \ldots)\,\mathrm{d}x + H_m(\psi(\beta), \ldots) - H_m(\psi(\alpha), \ldots). \end{aligned} \tag{6.3}$$

All integrals here are ordinary convergent integrals. For instance $H_4 = -2u_0^3 + 4u_0u_1 - u_2$, so

$$\int_\alpha^\beta \left\{ \Gamma(x)(\Gamma(x)^{-1})^{(4)} - \psi(x)^4 + \psi'(x)^2 \right\} \mathrm{d}x = \left[-2\psi^3 + 4\psi\psi' - \psi'' \right]_\alpha^\beta. \tag{6.4}$$

The supplied rational JSON certificates give A_m, N_m, H_m explicitly for weights $4, 8, 11, 14, 18$; the algorithms work at every finite weight.

At positive integer endpoints the entries are harmonic-tail polynomials:

$$\begin{aligned}\psi(N+1) &= H_N - \gamma,\\ \psi^{(j)}(N+1) &= (-1)^{j+1} j!(\zeta(j+1) - H_N^{(j+1)}), \qquad j \geq 1,\end{aligned} \tag{6.5}$$

where $H_N^{(q)} = \sum_{n=1}^{N} n^{-q}$. These are the standard polygamma recurrence and zeta specialization [7]. Equations (1.7), (6.2), and (6.5) therefore specify all endpoint constants in (6.3) by finite rational operations in harmonic numbers and zeta values.

Fixed-coordinate finite parts. If the lower endpoint is zero, retain the coordinate x and define the finite part by the constant term of the cutoff expansion after removing negative powers and logarithmic divergences. Since $H_m(\psi(x), \dots)$ has a Laurent expansion at zero, the corresponding identity is

$$\mathrm{FP}\int_0^{\beta} (A_m - N_m)(\psi(x), \dots)\,\mathrm{d}x = H_m(\psi(\beta), \dots) - [x^0]H_m(\psi(x), \dots). \tag{6.6}$$

The last term is generally not zero. This is the same endpoint convention as in the manuscript and predecessor [1, 3]; no coordinate-invariant finite-part assertion is intended.

What this does not imply. The theorem evaluates a particular differential-polynomial combination and its primitive. It does not turn each remaining normal monomial into a total derivative. For example, functional monomials were chosen precisely as a complement to total derivatives. Claims about individual integrals require additional analytic identities, such as the transforms in the next sections.

7 Beta logarithmic moments and generalized harmonic numbers

The quadratic-background coefficients naturally introduce the weights

$$P_{r,b}(v) = (1-v^2)^r(1+3v^2)^b, \qquad r, b \in \mathbb{Z}_{\geq 0}. \tag{7.1}$$

Their ordinary logarithmic moments have a finite all-order expression. For a finite alphabet $\boldsymbol{x} = (x_0, \dots, x_r)$, let $h_m(\boldsymbol{x})$ be the complete homogeneous polynomial defined by

$$\prod_{\ell=0}^{r}(1-x_\ell t)^{-1} = \sum_{m\geq 0} h_m(\boldsymbol{x})t^m. \tag{7.2}$$

This notation includes $h_0 = 1$.

Proposition 7.1 (Every logarithmic beta moment)**.** *For $\Re k > -1$, $m \geq 0$, and $r, b \geq 0$,*

$$\begin{aligned}&\int_0^1 v^k P_{r,b}(v)(-\log v)^m\,\mathrm{d}v\\ &= m!\,2^r r! \sum_{j=0}^{b}\binom{b}{j}3^j \frac{h_m\big((k+2j+1)^{-1}, \dots, (k+2j+2r+1)^{-1}\big)}{\prod_{\ell=0}^{r}(k+2j+2\ell+1)}.\end{aligned} \tag{7.3}$$

The dots in the alphabet mean spacing by two in the denominator.

Proof. Expand $(1+3v^2)^b$ and apply (1.9). Repeated k-differentiation is justified uniformly on compact subsets of $\Re k > -1$ by $v^{\Re k}|\log v|^m$ domination. Differentiating a product of reciprocal linear factors gives $(-1)^m m!$ times that product times its complete homogeneous polynomial in the reciprocals. □

The power sums of the alphabet in (7.3) are polygamma differences. With $x = (k+2j+1)/2$ and $q \geq 1$,

$$\sum_{\ell=0}^{r} \frac{1}{(k+2j+2\ell+1)^q} = \frac{(-1)^{q-1}}{2^q(q-1)!} \left\{ \psi^{(q-1)}(x+r+1) - \psi^{(q-1)}(x) \right\}. \tag{7.4}$$

For $k = 2n$, $n \geq 0$, the same sum is

$$H^{(q)}_{2(n+j+r+1)} - H^{(q)}_{2(n+j)} - 2^{-q} \left\{ H^{(q)}_{n+j+r+1} - H^{(q)}_{n+j} \right\}. \tag{7.5}$$

Together with the Newton identities for h_m, these formulas express all moments in generalized harmonic numbers at integer parameters. For example $h_1 = p_1$, $h_2 = (p_1^2+p_2)/2$, and $h_3 = (p_1^3+3p_1p_2+2p_3)/6$, where p_q denotes the power sum. The analytic statements are classical beta differentiation; their role here is to turn the new coefficient weights into explicit harmonic and polygamma identities at every logarithmic order.

8 Finite Hurwitz endpoint formulas for the coefficient weights

The universal formula supplies rational polynomial weights, not just rational coefficients. Their special-function transforms can be computed by a common finite endpoint calculus. Polynomial-weight Hurwitz integration is classical; see Espinosa–Moll [5, 6] and the standard formulas in [8]. Here we give a self-contained version and apply it to the weights arising from all Gamma sectors.

Let P be an even polynomial of degree D, vanishing at ± 1 to order at least $r \geq 1$. Every nonzero weight P_M in (1.6) has this property with $r = d$. For real $A > B > 0$, define

$$I_P(s; A, B) = \int_{-1}^{1} P(v)\zeta(s, A+Bv)\,\mathrm{d}v, \qquad \mu_P = \int_{-1}^{1} P(v)\,\mathrm{d}v. \tag{8.1}$$

Unlike the average notation in the coefficient theorem, I_P uses the *full* integral. We keep this normalization visible in all residues and examples.

Theorem 8.1 (Finite endpoints and the Mellin bridge)**.** *With $d_j = P^{(j)}(1)$,*

$$\boxed{I_P(s; A, B) = \sum_{j=r}^{D} \frac{(-1)^j d_j}{B^{j+1}(1-s)_{j+1}} \left\{ \zeta(s-j-1, A+B) - (-1)^j \zeta(s-j-1, A-B) \right\}.} \tag{8.2}$$

Here $(a)_n = a(a+1)\cdots(a+n-1)$ is the rising factorial. The equality is meromorphic in s. The only possible pole of the whole expression is $s = 1$, with residue μ_P. For $\Re s > 1$ it also equals the ordinary convergent Mellin integral

$$I_P(s; A, B) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}e^{-At}}{1-e^{-t}} \left(\int_{-1}^{1} P(v) e^{-Btv}\,\mathrm{d}v \right) \mathrm{d}t. \tag{8.3}$$

Proof. The standard parameter derivative is $\partial_a \zeta(s,a) = -s\zeta(s+1,a)$ [8]. Consequently an antiderivative in v of $\zeta(s, A+Bv)$ is $\zeta(s-1, A+Bv)/(B(1-s))$. Repeated integration by parts terminates after degree D and gives (8.2), because $P^{(j)}(-1) = (-1)^j d_j$ and $d_j = 0$ for $j < r$. Initially exclude the finite exceptional set where a displayed intermediate expression is singular; meromorphic continuation gives the identity elsewhere.

The Hurwitz zeta function, uniformly for its parameter in the compact positive interval $[A-B, A+B]$, has only the simple pole at $s=1$ with residue 1. Integrating its holomorphic pole-subtracted germ proves the assertion about the whole sum. This also shows why apparent singularities of individual endpoint terms must be combined before evaluation.

Finally insert the standard Mellin representation of $\zeta(s,a)$. Near zero the absolute integrand is $O(t^{\Re s-2})$; at infinity it is dominated by a polynomial times $e^{-(A-B)t}$. Fubini's theorem proves (8.3) in its stated domain. □

At $s \neq 1$, the compact integral (8.1) is an ordinary integral even when the infinite Mellin integral does not converge. The latter must not be used outside its convergence domain without an explicit continuation or subtraction procedure.

For $q \geq 1$, the polygamma and Gamma specializations are

$$\int_{-1}^{1} P(v)\psi^{(q)}(A+Bv)\,\mathrm{d}v = (-1)^{q+1} q!\, I_P(q+1; A, B), \tag{8.4}$$

$$\int_{-1}^{1} P(v)\log\Gamma(A+Bv)\,\mathrm{d}v = \partial_s I_P(s; A, B)|_{s=0} + \frac{\mu_P}{2}\log(2\pi). \tag{8.5}$$

These follow from $\psi^{(q)}(a) = (-1)^{q+1} q! \zeta(q+1,a)$ and $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ [7, 8]. Differentiation in s, A, B is legitimate on compact subsets away from genuine poles and away from the parameter boundary $A = B$.

9 Every Stieltjes order and every anchored parameter primitive

Use the Laurent convention

$$\zeta(1+\varepsilon, a) = \frac{1}{\varepsilon} + \sum_{m\geq 0} \frac{(-1)^m \gamma_m(a)}{m!}\varepsilon^m, \qquad \gamma_0(a) = -\psi(a). \tag{9.1}$$

For $j \geq 0$, define complete-harmonic coefficients by

$$\prod_{\ell=1}^{j} (1-\varepsilon/\ell)^{-1} = \sum_{n\geq 0} h_n^{(j)} \varepsilon^n. \tag{9.2}$$

For the empty product, $h_0^{(0)} = 1$ and $h_n^{(0)} = 0$ for $n > 0$. For example

$$h_1^{(j)} = H_j, \qquad h_2^{(j)} = \frac{H_j^2 + H_j^{(2)}}{2}, \qquad h_3^{(j)} = \frac{H_j^3 + 3H_j H_j^{(2)} + 2H_j^{(3)}}{6}.$$

Write $\zeta^{[\ell]}(s,a) = \partial_s^\ell \zeta(s,a)$ for spectral derivatives, to distinguish them from parameter derivatives.

Theorem 9.1 (All-order Stieltjes endpoint formula)**.** *For P, A, B as in Theorem 8.1, set*

$$c_j(P,B) = \frac{(-1)^j P^{(j)}(1)}{B^{j+1} j!}. \tag{9.3}$$

Then for every $m \geq 0$,

$$\int_{-1}^{1} P(v)\gamma_m(A+Bv)\,\mathrm{d}v$$
$$= (-1)^{m+1}m!\sum_{j=r}^{D} c_j(P,B)\sum_{\ell=0}^{m+1}\frac{h_{m+1-\ell}^{(j)}}{\ell!}\left\{\zeta^{[\ell]}(-j,A+B)-(-1)^j\zeta^{[\ell]}(-j,A-B)\right\}. \tag{9.4}$$

In particular, the required endpoint spectral derivatives have orders at most $m+1$, at the finitely many nonpositive integers $-j$.

Proof. At $s = 1+\varepsilon$,

$$\frac{1}{(1-s)_{j+1}} = -\frac{1}{\varepsilon j!}\prod_{\ell=1}^{j}(1-\varepsilon/\ell)^{-1}. \tag{9.5}$$

Insert this in (8.2), Taylor-expand the two zeta values, and compare the coefficient of ε^m with (9.1). The finite endpoint sum permits coefficient extraction term by term. On the compact parameter interval, the pole-subtracted Hurwitz germ is jointly holomorphic, so its Laurent coefficients are the ordinary integrals of γ_m. □

Thus the Stieltjes calculation is an explicit convolution with complete harmonic numbers, not an unspecified instruction to "differentiate a regularized expression." The residue identity, obtained at the previous coefficient, is

$$-\sum_{j=r}^{D} c_j(P,B)\left\{\zeta(-j,A+B)-(-1)^j\zeta(-j,A-B)\right\} = \mu_P. \tag{9.6}$$

It gives a useful exact Bernoulli-polynomial check of all endpoint normalizations.

9.1 Arbitrary orders of antiderivation in the center

For integers $m \geq 0$, $d \geq 1$, define

$$\mathcal{B}_{P;m,d}(A;B) = \frac{(-1)^{m+1}m!}{(d-1)!}\sum_{\ell=0}^{m+1}\frac{h_{m+1-\ell}^{(d-1)}}{\ell!}\,\partial_s^{\ell} I_P(s;A,B)\Big|_{s=1-d}. \tag{9.7}$$

Theorem 9.2 (All anchored primitives)**.** *For $A > B > 0$,*

$$\partial_A^d \mathcal{B}_{P;m,d}(A;B) = \int_{-1}^{1} P(v)\gamma_m(A+Bv)\,\mathrm{d}v. \tag{9.8}$$

Fix an anchor $A_0 > B$. The unique d-fold primitive whose first d initial data vanish at A_0 is

$$\mathcal{B}_{P;m,d}(A;B) - \sum_{h=0}^{d-1}\frac{(A-A_0)^h}{h!}\partial_A^h \mathcal{B}_{P;m,d}(A_0;B). \tag{9.9}$$

Proof. The repeated Hurwitz parameter derivative gives, meromorphically in s,

$$\partial_A^d\left\{\frac{I_P(s-d;A,B)}{(1-s)_d}\right\} = I_P(s;A,B).$$

Set $s = 1 + \varepsilon$ and use

$$\frac{1}{(-\varepsilon)_d} = -\frac{1}{\varepsilon(d-1)!}\prod_{\ell=1}^{d-1}(1-\varepsilon/\ell)^{-1}.$$

The coefficient of ε^m, multiplied by $(-1)^m m!$, is (9.7). Holomorphic dependence on A in $\Re A > B$ permits commutation of the derivative and coefficient extraction. Subtracting the Taylor polynomial supplies the stated initial conditions. The difference of any two such primitives is a polynomial in A of degree at most $d-1$, hence is zero. □

For $m = 0, d = 1$, the anchored primitive simplifies to the ordinary Gamma ratio integral

$$-\int_{-1}^{1} P(v)\log\frac{\Gamma(A+Bv)}{\Gamma(A_0+Bv)}\,\mathrm{d}v. \tag{9.10}$$

Equations (8.2) and (9.7) also give a finite endpoint expression for every higher primitive. No unchosen polynomial integration constant is hidden in the notation.

10 Lerch deformations, polylogarithms, and Fourier endpoint identities

10.1 A finite polylogarithmic endpoint calculus

For $|\xi| < 1$, the Lerch series is

$$\Phi(\xi, s, a) = \sum_{n\geq 0}\frac{\xi^n}{(n+a)^s}, \qquad a > 0,$$

where positive real denominators use the real logarithm [10]. It converges normally in s on every compact set. Define

$$L_P(s; A, B; \xi) = \int_{-1}^{1} P(v)\Phi(\xi, s, A+Bv)\,\mathrm{d}v. \tag{10.1}$$

The same finite integration-by-parts proof gives

$$L_P(s; A, B; \xi) = \sum_{j=r}^{D}\frac{(-1)^j d_j}{B^{j+1}(1-s)_{j+1}}\Big\{\Phi(\xi, s-j-1, A+B) - (-1)^j\Phi(\xi, s-j-1, A-B)\Big\}. \tag{10.2}$$

The whole expression is entire in s; every apparent exceptional value is obtained by joint continuation. In $\Re s > 0$ its ordinary Mellin representation replaces $1-e^{-t}$ in (8.3) by $1-\xi e^{-t}$. The weaker lower-end restriction follows because the new denominator is nonzero at $t = 0$.

If $A \pm B$ are positive integers and $0 < |\xi| < 1$, the elementary identity

$$\Phi(\xi, s, N) = \xi^{-N}\left\{\mathrm{Li}_s(\xi) - \sum_{n=1}^{N-1}\frac{\xi^n}{n^s}\right\} \tag{10.3}$$

turns (10.2) into a finite combination of polylogarithms at shifted orders and finite harmonic-type sums. Every spectral derivative and every center primitive follows from the same identities. The apparent singularity at $\xi = 0$ in (10.3) is removable after the displayed difference is formed. At $\xi = 1$, passage to Hurwitz zeta must respect the genuine pole and the original convergence domain; the entire-in-s statement for $|\xi| < 1$ is not uniform across that boundary.

10.2 A finite zeta formula at every complex Fourier order

The same endpoint polynomial also evaluates a Fourier-polylogarithm integral without a Hurwitz parameter.

Proposition 10.1 (Fourier endpoint identity)**.** *Let P be an even polynomial of degree D, with $P(\pm 1) = 0$. For $\Re s > 1$,*

$$\boxed{\int_{-1}^{1} P(v)\,\mathrm{Li}_s(e^{2\pi \mathrm{i}v})\,\mathrm{d}v = 2\sum_{\ell=1}^{D/2} \frac{(-1)^{\ell+1}P^{(2\ell-1)}(1)}{(2\pi)^{2\ell}}\zeta(s+2\ell).} \tag{10.4}$$

The integer upper limit is harmless because D is even; zero endpoint derivatives may be omitted. The identity may be differentiated in s any fixed number of times within this half-plane.

Proof. For $n \geq 1$, repeated integration by parts gives the exact terminating Fourier coefficient

$$\int_{-1}^{1} P(v)e^{2\pi \mathrm{i}nv}\,\mathrm{d}v = 2\sum_{\ell=1}^{D/2} \frac{(-1)^{\ell+1}P^{(2\ell-1)}(1)}{(2\pi n)^{2\ell}}.$$

The even endpoint derivatives cancel between 1 and -1, and the last constant derivative has zero integral over the integer number of periods. Now use the absolutely uniformly convergent unit-circle series for Li_s when $\Re s > 1$ [11]. Summing n^{-s} times the finite Fourier coefficient gives (10.4). Spectral derivatives introduce only powers of $\log n$, preserving normal convergence on smaller compact subsets of the half-plane. □

At $s = 1$, the ordinary integrable logarithmic boundary value gives

$$\int_{-1}^{1} P(v)\log(2|\sin \pi v|)\,\mathrm{d}v = -2\sum_{\ell=1}^{D/2} \frac{(-1)^{\ell+1}P^{(2\ell-1)}(1)}{(2\pi)^{2\ell}}\zeta(2\ell+1). \tag{10.5}$$

One rigorous route is to insert a radial factor $0 < \rho < 1$ into the Fourier series and use $\mathrm{Li}_1(z) = -\log(1-z)$. The real part converges in L^1 to $-\log(2|\sin \pi v|)$: near each integer the absolute logarithm is dominated by an integrable constant plus $|\log|v-n||$, uniformly for ρ sufficiently close to one. The integrated Fourier series is absolutely summable by its explicit coefficients. The imaginary part cancels by parity. No unregulated ordinary-integral statement at $s = 0$ or negative orders is inferred; such boundary values can involve distributional terms.

11 A degree-eight bridge: four explicit evaluations

The first genuinely mixed-square coefficient $u_2u_3^2$ produces

$$P(v) = P_{3,1}(v) = (1-v^2)^3(1+3v^2) = 1 - 6v^4 + 8v^6 - 3v^8. \tag{11.1}$$

On the interval $1 \leq x \leq 3$ put

$$W(x) = P(x-2) = (x-1)^3(3-x)^3(1+3(x-2)^2). \tag{11.2}$$

Its mass is $\int_1^3 W(x)\,\mathrm{d}x = 128/105$. The endpoint derivatives provide a compact exact certificate:

j	3	4	5	6	7	8
$P^{(j)}(1)$	-192	-2304	-14400	-54720	-120960	-120960
$(-1)^j P^{(j)}(1)/j!$	32	-96	120	-76	24	-3

Here $A = 2$, $B = 1$ in the compact Hurwitz transform. Integer translation reduces its finite endpoint formula to

$$I_P(s; 2, 1) = \sum_{j=3}^{8} \frac{(-1)^j P^{(j)}(1)}{(1-s)_{j+1}} \left\{ (1 - (-1)^j)\zeta(s-j-1) - 1 - 2^{j+1-s} \right\}. \tag{11.3}$$

This one meromorphic identity proves the following evaluations.

11.1 Trigamma and the zeroth Stieltjes constant

$$\boxed{\int_1^3 W(x)\psi'(x)\,\mathrm{d}x = -\frac{32}{35} - \frac{48}{\pi^2}\zeta(3) + \frac{900}{\pi^4}\zeta(5) - \frac{1890}{\pi^6}\zeta(7).} \tag{11.4}$$

For completeness, at $s = 2 + \varepsilon$,

$$\frac{1}{(1-s)_{j+1}} = \frac{1}{\varepsilon(j-1)!} \left\{ 1 + (H_{j-1} - 1)\varepsilon + O(\varepsilon^2) \right\}.$$

The table shows that the total residue in (11.3) is zero. Its elementary finite part is $-32/35$; all $\log 2$ terms cancel. For odd $j = 2n + 1$, the zeta derivative is reduced by the standard functional-equation consequence [9]

$$\zeta'(-2n) = \frac{(-1)^n (2n)!}{2(2\pi)^{2n}} \zeta(2n+1).$$

This gives (11.4) with no appeal to a numerical integer-relation guess.

The constant term at the genuine pole $s = 1$ is

$$\boxed{\int_1^3 W(x)\gamma_0(x)\,\mathrm{d}x = -\frac{547}{1800} - 64\zeta'(-3) - 240\zeta'(-5) - 48\zeta'(-7).} \tag{11.5}$$

Since $\gamma_0 = -\psi$, this is also an explicit weighted digamma integral. The derivatives in this and the next display are derivatives of the ordinary Riemann zeta function in its argument.

11.2 An explicit first-Stieltjes integral

The next coefficient of (11.3) gives

$$\begin{aligned} \int_1^3 W(x)\gamma_1(x)\,\mathrm{d}x = \frac{3499687}{10584000} - \frac{32}{21}\log 2 + \frac{352}{3}\zeta'(-3) + 548\zeta'(-5) + \frac{4356}{35}\zeta'(-7) \\ + 32\zeta''(-3) + 120\zeta''(-5) + 24\zeta''(-7). \end{aligned} \tag{11.6}$$

To check all terms without numerical approximation, put $c_j = (-1)^j P^{(j)}(1)/j!$ and expand

$$B_j(\varepsilon) = (1 - (-1)^j)\zeta(-j + \varepsilon) - 1 - 2^j e^{-\varepsilon \log 2}.$$

Then $I_P(1+\varepsilon;2,1)$ is

$$-\frac{1}{\varepsilon}\sum_{j=3}^{8} c_j B_j(\varepsilon)\left\{1+H_j\varepsilon+\tfrac{1}{2}(H_j^2+H_j^{(2)})\varepsilon^2+\cdots\right\}.$$

The coefficient of ε is the negative of the right side of (11.6). This proves the displayed rational, logarithmic, first-derivative, and second-derivative coefficients simultaneously. Theorem 9.1 gives every subsequent γ_m integral by the same finite rule.

11.3 A function-valued polylogarithm identity and a log-sine value

For $\Re s > 1$, Proposition 10.1 specializes to

$$\boxed{\int_1^3 W(x)\,\mathrm{Li}_s(e^{2\pi i x})\,\mathrm{d}x = \frac{24}{\pi^4}\zeta(s+4)-\frac{450}{\pi^6}\zeta(s+6)+\frac{945}{\pi^8}\zeta(s+8).} \tag{11.7}$$

The unit-circle polylogarithm here is defined by its convergent series or equivalently its radial boundary value from the disk. This is an identity in a complex order, not only a list of special integer values. All fixed spectral derivative orders obey the differentiated identity. The logarithmic endpoint specialization is

$$\boxed{\int_1^3 W(x)\log(2|\sin\pi x|)\,\mathrm{d}x = -\frac{24}{\pi^4}\zeta(5)+\frac{450}{\pi^6}\zeta(7)-\frac{945}{\pi^8}\zeta(9).} \tag{11.8}$$

There is an interior logarithmic singularity at $x=2$, but it is integrable. No finite-part convention is used in any of (11.4)–(11.8). The finite Lerch formula (10.2) gives a second, disk-valued polylogarithmic deformation of the same weight, independently of the Fourier specialization.

12 Source audit, corrections, and reproducible verification

12.1 Provenance and the exact advance

The research was based on targeted reads of the canonical manuscript metadata and relevant normal-form material, the harmonic-tail Gamma continuation [2], and the affine-polarization source [3]. The incoming directory was inspected at the metadata level. The affine article was retrieved as a full TeX file from the user's file library; byte identity with a similarly named incoming ZIP was not established. Other incoming archives were not exhaustively unpacked. This is a targeted continuation, not a complete audit of the large and actively changing repository.

The inspected affine TeX file is named `ProveIt_Affine_Gamma_Polarization_2026-10-11.tex`. Its SHA-256 fingerprint is

`36a06cf61ab08597c5526c3ced33af296fb`
`536466974211073769ee41218c751`

The file itself records the older repository metadata anchor `92ee3aaf3087c0e0c29b9ed952e777de5d6accd0`. That is the predecessor's provenance, not a claim that every live repository read in this session came from that revision. Observed search results moved through several revisions; the final additional search results carried `af91e74dfa841331c51e867542345a3643ee0bf0`. The source manifest in the package records this distinction.

The contributions are separated as follows. The functional complement, centering, Appell factor, and the affine kernel are inherited and explicitly proved again for normalization. The polynomial-background transport, its highest-variation use to determine *all* normal coefficients, the universal formula, the sufficient support condition, and the highest-index sign law constitute the present coefficient advance relative to the inspected sources. The finite u_2 sum and all mixed-square formulas are consequences, not separate conjectural extensions. The higher-multiplicity sectors are included in the theorem.

Polynomial-weight Hurwitz integration, beta differentiation, and the Lerch–polylogarithm connection are established methods [5, 6, 8, 10, 11]. The analytic contribution is their explicit application and all-order normalization for the weights supplied by the universal Gamma formula, including the worked degree-eight identities. Neither the general method nor historical priority for each isolated definite integral is claimed as new. Targeted repository and public-source searches did not supply the universal formula, but those searches do not establish exhaustive historical priority.

12.2 Corrections and safeguards

A definite bibliographic correction. The affine report's bibliography gives pages 456–473 for Guo, Regensburger, and Rosenkranz [4]. The authors' arXiv record gives *Journal of Pure and Applied Algebra* **218** (2014), 456–471, with the same DOI. The page range should therefore be changed to 456–471. This does not affect any mathematical argument.

A repository editorial duplication. The initially inspected canonical manuscript README, with blob identity `dc0fa26d0e5c2ede56df37c5640c7319681dfd25`, repeated the paragraph about the complete 1,032-input first-letter X family, rank 1,008, and kernel 24. Remove the second copy if that duplication remains in the integration target. This observation is tied to the inspected blob, not a claim about every later version of the moving default branch.

No substantive predecessor error established. The coefficient formulas used from the two direct predecessors agree with the present formula in their overlapping sectors. We did not identify a substantive mathematical error in those inspected formulas. The following are implementation safeguards, not retrospective accusations: reduce before truncation; do not multiply projected normal forms as though the projection were an algebra homomorphism; keep the $e^{z/24}$ factor in the quadratic-background transport; combine removable endpoint singularities before evaluation; and retain the primitive's lower-end constant in fixed-coordinate finite parts.

12.3 Independent exact checks

The package contains two different algorithms. The coefficient module `verification/gamma_coefficients.py` uses only polynomial arithmetic and the universal beta formula. The reference module `verification/independent_ibp.py` generates A_m by (1.2) and applies the terminating local reduction (2.2); it does not call the coefficient formula.

The exact run checks every functional exponent slot through weight 32, including those predicted to vanish. The original Bell polynomial has nonzero coefficient at every partition of its weight, so its functional slots, together with the reference normal form's support, enumerate the whole required test set. The recorded results are:

Check category	Count
Independent rational coefficient comparisons, weights 0–32	8,350
Exact support and sign checks on positive highest indices	8,317
Full identities $A_m = N_m + \mathcal{D}H_m$, weights 0–28	29
Euler-homotopy versus recursive primitive, weights 0–18	19
Appell derivative identities, weights 1–32	32
Mixed-square finite-sum comparisons	1,684
Deliberately corrupted certificates rejected	2
Exact analytic endpoint-algebra assertions	20

The two corrupted certificates change the displayed weight-eleven and cubic weight-fourteen coefficients while retaining their independently computed primitives; the polynomial identity fails as it should. The exact endpoint assertions include Bernoulli-polynomial evaluations at thirteen nonpositive spectral orders, the residue, and the rational and harmonic coefficient algebra of the worked examples. The supplied CSV contains all 8,350 coefficient slots, including zeros. The JSON certificates contain complete A_m, N_m, H_m data at the five selected weights, not just numerical evaluations.

12.4 Numerical diagnostics and their limits

The analytic script was run with 65 decimal digits using Python 3.13.5, mpmath 1.3.0, and SymPy 1.14.0. All 20 diagnostics passed their declared local tolerances. They include real and complex Hurwitz orders, trigamma, γ_0, γ_1, a log-sine integral, a finite polylogarithmic endpoint formula, mixed beta moments, four logarithmic orders, an anchored Gamma primitive, and an ordinary Gamma endpoint identity.

For independent quadrature values L, R, the recorded residual is $|L - R|/(1 + |L| + |R|)$. The largest recorded residual is about 4.82×10^{-34}, in the refinement comparison between 32- and 48-point Gauss–Legendre quadrature for the γ_1 integral; its declared tolerance is 10^{-31}. The finer quadrature versus the finite endpoint value differs by about 2.68×10^{-52} in the same scaled measure. These are floating-point diagnostics, not rigorous interval enclosures, and 65-digit working precision does not mean that every integral has 65 certified digits.

For orientation, the four ordinary integrals in (11.4), (11.5), (11.6), and (11.8) begin

$$0.83786519860377187790, \qquad -0.475626291891128162\,16,$$
$$-0.11146849703321255941, \qquad 0.11670482124545205075,$$

respectively. The finite identities and their proofs, rather than these decimal values, are the research statements.

Replay and integration. From the package root, run

```
python verification/check_exact.py
python verification/check_analytic.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact coefficient checks need only the Python standard library. The analytic checks require the two packages listed above. The archive includes an additive integration plan and a compact theorem fragment with unique `ubg:` labels. No connected repository files were modified. The result has not been formalized in a proof assistant.

13 Further research questions after the coefficient theorem

The all-monomial theorem changes the frontier: finding the next mixed Gamma coefficient is no longer an open problem for this hierarchy. The following targets concern genuinely additional structure rather than an already solved sector.

1. Exact denominators and arithmetic normalization. Determine the least positive integer D_m for which $D_m N_m$ has integer coefficients, and characterize the denominator of an individual coefficient from its exponent vector. The first cubic example $-3185/3$ rules out a blanket integrality statement. The positive finite sum (1.10) gives a concrete starting point for prime-adic cancellation identities. A satisfactory answer should specify exact valuations or divisibility laws, not merely denominator bounds.

2. Analytic summation of the full normal-form series. For an analytic function $u_0(x)$, determine when the formal infinite sum underlying (5.2) converges after all jet substitutions, and identify its analytic value. The theorem proves entire dependence for every *fixed finite jet*, not convergence of the infinitely many jets of $\psi(x)$. It would be particularly useful to obtain an exact integral representation on a specified (x, t) domain and the corresponding summed endpoint primitive.

3. General centered profiles and products of Gamma ratios. Replace $K_t(\mathcal{D})$ by an even profile with leading term t, or consider finite products of centered Gamma ratios. The affine predecessor already supplies square symbols for general symmetric profiles [3]. Differential superposition still provides a route to polynomial backgrounds. Determine which profiles admit an equally simple full-coefficient measure, and give necessary and sufficient conditions for the exact sign law to survive. Positivity of the Gamma beta kernel is a special input, not a formal consequence for every profile.

4. Arithmetic selection by coefficient weights. The predecessor's Bernoulli selectors diagonalize certain polynomial Hurwitz integrals into single shifted zeta terms [3]. Classify the universal weights P_M that lie in one selector line, or in a prescribed low-dimensional selector span. This asks which actual Gamma coefficient patterns produce a small number of zeta values, rather than redoing the existing classification of arbitrary even polynomial weights.

5. Exact values of individual nonlinear polygamma integrals. The universal normal form supplies all linear relations arising from this Gamma hierarchy, but it does not evaluate every surviving functional integral. A concrete target is to identify the natural function algebra for $\int_1^2 \psi'(x)(\psi'''(x))^3\,\mathrm{d}x$, or for a parameterized version with independently shifted arguments. Investigate a finite reduction to multiple Hurwitz or multiple-polylogarithmic quantities with fixed branches, and distinguish an actual evaluation from a relation in which new unevaluated integrals remain.

6. Endpoint collisions in the Stieltjes and Lerch transforms. The present formulas assume $A > B > 0$. Continue the exact endpoint identities as $A \downarrow B$, keeping a specified coordinate and subtraction convention. Which weights kill every local singular term, and which leave a nonzero contact contribution? The ordinary log-sine limit at order one does not settle the distributional limits at nonpositive polylogarithm orders. The repository's boundary-contact work should be used rather than silently identifying the two regimes.

7. A proof-carrying compiler. Formalize the terminating functional decomposition, weighted quadratic symbol, differential superposition, and highest-variation extraction. The theorem splits naturally into finite polynomial identities and coefficientwise formal-series arguments. A first practical milestone is a proof-assistant checker for the supplied rational certificates; a separate milestone is a formal all-order proof. Certified interval versions of the analytic diagnostics would add numerical evidence but would not substitute for either formal theorem.

8. Preserve the original period and tail-rank targets. The exact Gaussian S_6 and revised S_8 candidates still require branch-controlled polylogarithmic identities with their complete lower-weight constants. The general unequal harmonic-tail moment-prefix rank question still requires a genuine determinant or independence argument. The present coefficient theorem gives additional exact identities and endpoint tools, but no implication resolving those separate questions is claimed.

Conclusion

The normal-form coefficient problem has a uniform answer: its highest quadratic variation on a polynomial background is a compact exponential beta integral, and two integrations in the highest jet recover every multiplicity. This yields a finite rational formula for every functional coefficient, a complete support criterion, and a uniform sign law. Keeping the primitive gives exact Gamma–harmonic endpoint identities; retaining the polynomial weight gives finite zeta, Stieltjes, and polylogarithmic transforms. The distinctions between formal reduction, ordinary integration, joint meromorphic continuation, and numerical verification remain explicit throughout.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*. Canonical manuscript, especially the normal-form material and chapters `07-ibp-normal-forms.tex` and `07-differential-polynomial-moments.tex`. Targeted repository reads, 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Polylogarithms/docs/manuscript. The incoming-directory inspection was at https://github.com/VladimirReshetnikov/ProveIt/tree/main/docs/incoming.

[2] *Exact Harmonic-Tail Reconstruction and Variational Gamma Identities*. ProveIt working report, 11 October 2026. Especially `sections/04_gamma_sectors.tex`; article blob identity `445e67f0cd60313113387d52d44d4ddc6eeaf244`. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/harmonic-tail-gamma.

[3] *Affine Polarization of Gamma Normal Forms: All derivative-square coefficients, polylogarithmic kernels, and Stieltjes averages*. Research continuation prepared for Vladimir Reshetnikov, 11 October 2026. Full TeX source retrieved from the user's file library as `ProveIt_Affine_Gamma_Polarization_2026-10-11.tex`. Sections 2–3 and 11 are the direct normal-form input and open-question source. The exact source fingerprint and inspection limits appear in Section 12 and the package manifest.

[4] Li Guo, Georg Regensburger, and Markus Rosenkranz, *On integro-differential algebras*. Journal of Pure and Applied Algebra **218** (2014), 456–471. https://doi.org/10.1016/j.jpaa.2013.06.015. Author preprint and journal-reference

record: https://arxiv.org/abs/1212.0266. Section 4.4.1 supplies the classical functional-monomial construction.

[5] Olivier R. Espinosa and Victor H. Moll, *On some definite integrals involving the Hurwitz zeta function*. Author preprint, 2000. https://arxiv.org/abs/math/0012078. Cited for the established polynomial-weight Hurwitz, Bernoulli, and log-Gamma integration programme.

[6] Olivier R. Espinosa and Victor H. Moll, *On some integrals involving the Hurwitz zeta function: part 2*. Author preprint, 2001. https://arxiv.org/abs/math/0107082. Cited for established antiderivative and generalized-polygamma methods.

[7] NIST Digital Library of Mathematical Functions, §5.15, *Polygamma Functions*. Consulted 11 October 2026. https://dlmf.nist.gov/5.15.

[8] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. Consulted 11 October 2026. https://dlmf.nist.gov/25.11. Definitions, parameter derivatives, integer shifts, Bernoulli values, Mellin representation, and log-Gamma specialization.

[9] NIST Digital Library of Mathematical Functions, §25.6, *Integer Arguments*. Consulted 11 October 2026. https://dlmf.nist.gov/25.6.

[10] NIST Digital Library of Mathematical Functions, §25.14, *Lerch's Transcendent*. Consulted 11 October 2026. https://dlmf.nist.gov/25.14.

[11] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. Consulted 11 October 2026. https://dlmf.nist.gov/25.12.